Note 9.2 First-Order Linear Equations

- **First-Order Linear Equations**
  It can be written in the form

$$\frac{dy}{dx} + P(x)y = Q(x)$$

  where P and Q are continuous functions of x. And the form is the linear equation's standard form.

Solving Linear Equations

- **Conclusion**
  To solve the linear equation in the standard form, multiply both sides by the integrating factor $v(x) = e^{\int P(x)dx}$ and integrate both sides.
- **Proof**
  Here is why multiplying by $v(x)$ works:

$$\begin{aligned}
\frac{dy}{dx} + P(x)y &= Q(x) \\
v(x)\frac{dy}{dx} + P(x)v(x)y &= v(x)Q(x) \\
\frac{d}{dx}(v(x) \cdot y) &= v(x)Q(x) \\
v(x) \cdot y &= \int v(x)Q(x)\,dx \\
y &= \frac{1}{v(x)}\int v(x)Q(x)\,dx
\end{aligned}$$

  Thus we call $v(x)$ an integrating factor.

Exercise Section 9.2

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- Question 29th
  For the Bernoulli differential equation in the form of

$$\frac{dy}{dx} + P(x)y = Q(x)y^n$$

  , if $n > 1$, w can substitute $u = y^{1-n}$ to transform it into the linear equation

$$\frac{du}{dx} + (1-n)P(x)u = (1-n)Q(x)$$